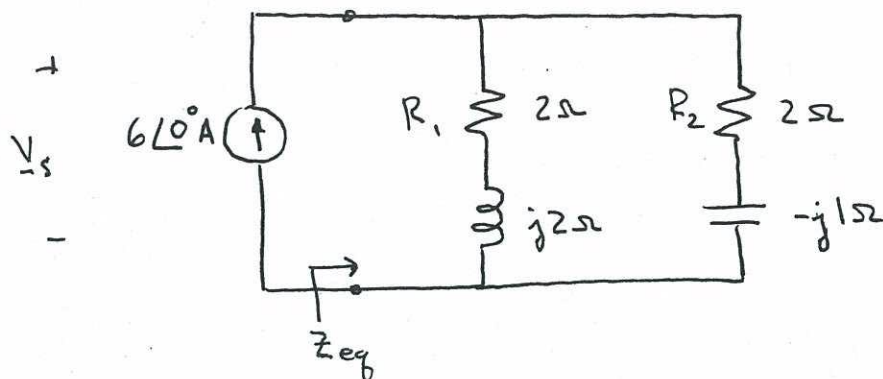


12.1



$$Z_{eq} = \frac{(2+j2)(2-j)}{2+j2+2-j} = \frac{6+j2}{4+j} = \frac{6.32 \angle 18.4^\circ}{4.12 \angle 14^\circ}$$

$$V_s = I_s \cdot Z_{eq} = 6 \angle 0^\circ \left[\frac{6.32 \angle 18.4^\circ}{4.12 \angle 14^\circ} \right] = 9.2 \angle 4.39^\circ \text{ V}$$

- Power supplied by source:

$$P_s = \frac{1}{2} (9.2)(6) \cos(4.39^\circ - 0^\circ) = \underline{\underline{27.52 \text{ W}}}$$

- Power absorbed by R_1 :

$$I_1 = \frac{6 \angle 0^\circ (2-j)}{4+j} = 3.26 \angle -40.6^\circ \text{ A} \Rightarrow P_{R1} = \frac{1}{2} (3.26^2) (2\Omega)$$

$$\underline{\underline{P_{R1} = 10.63 \text{ W}}}$$

- Power absorbed by R_2 :

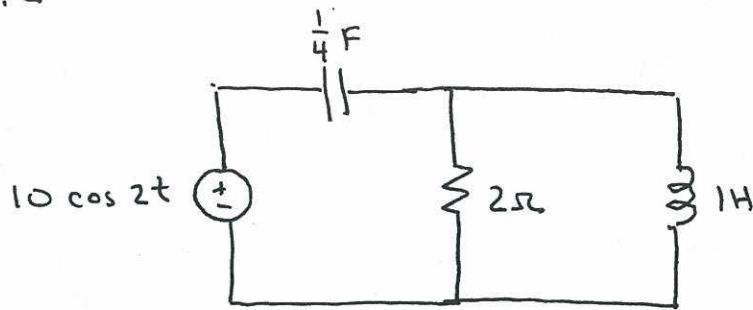
$$I_2 = \frac{6 \angle 0^\circ (2+j2)}{4+j1} = 4.12 \angle 30.96^\circ \text{ A} \Rightarrow P_{R2} = \frac{1}{2} (4.12)^2 (2\Omega)$$

$$\underline{\underline{P_{R2} = 16.9 \text{ W}}}$$

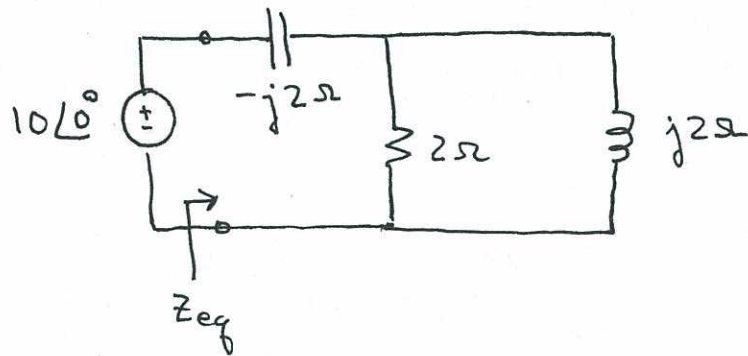
- No average power is absorbed or generated by the inductor or capacitor.

- Note: $P_{R1} + P_{R2} = P_s \quad \checkmark$

12.2



Frequency domain circuit:

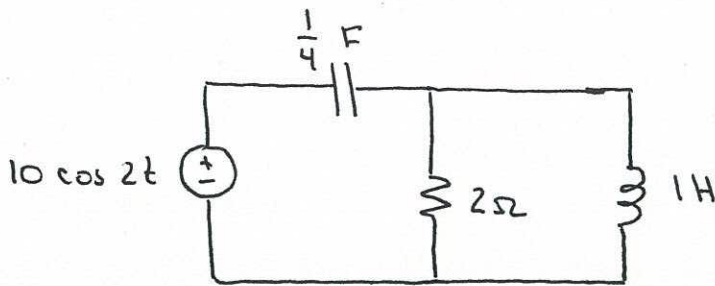


$$Z_{eq} = -j2\Omega + \frac{(2\Omega)(j2\Omega)}{(2+j2)\Omega} = (1-j)\Omega$$

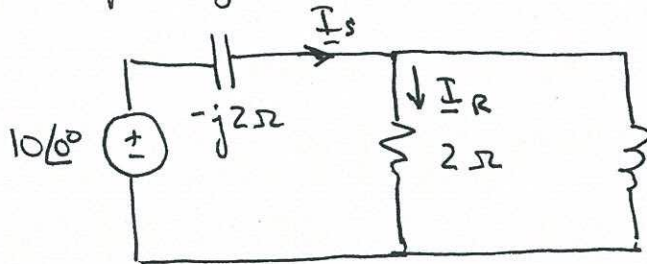
$$\underline{I}_s = \frac{10\angle 0^\circ}{1-j} = \frac{10}{\sqrt{2}} \angle 45^\circ$$

$$P_s = \frac{10 \left(\frac{10}{\sqrt{2}} \right)}{2} \cos(0^\circ - 45^\circ) \Rightarrow \underline{\underline{P_s = 25 \text{ W}}}$$

12.3



Frequency domain circuit:



From previous problem, $\underline{I}_s = \frac{10\angle 0^\circ}{1-j}$

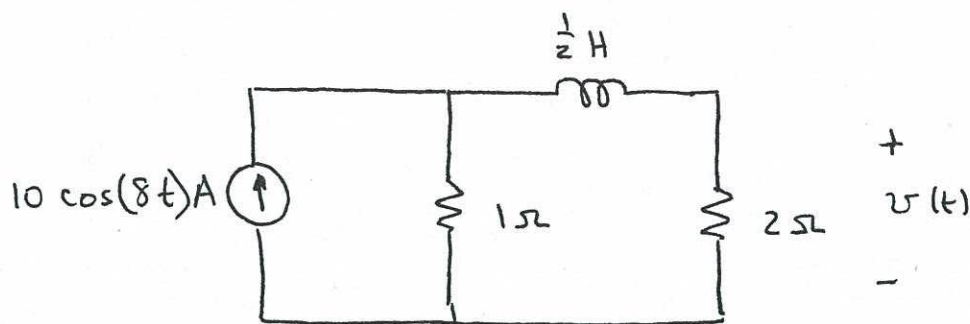
$$\underline{I}_R = \underbrace{\underline{I}_s \left(\frac{j2\Omega}{(2+j2)\Omega} \right)}_{\text{current divider}} = \underbrace{\frac{10\angle 0^\circ}{1-j}}_{\underline{I}_s} \left(\frac{j}{1+j} \right) = \frac{j10}{2}$$

$$\underline{I}_R = j5 \text{ A} \Rightarrow \underline{I}_R = 5\angle 90^\circ$$

$$P_R = \frac{\underline{I}_R^2}{2} \cdot R = \frac{5^2}{2} (2\Omega) \Rightarrow \underline{\underline{P_R = 25 \text{ W}}}$$

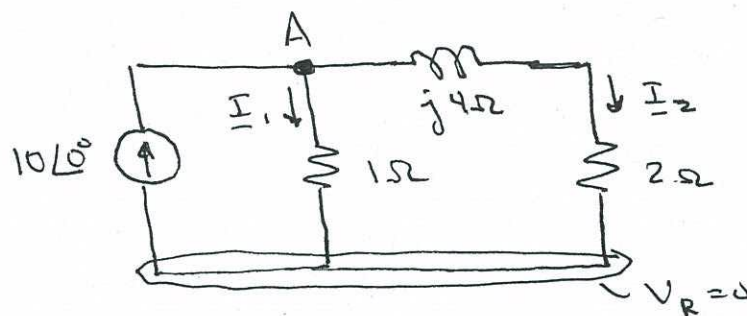
Note: We calculated the average power delivered by the source in the last problem. The power delivered by the source is equal to the power absorbed by the resistor.

12.4



(a) I'll determine $v(t)$ using several different approaches, to illustrate the flexibility of impedance analysis techniques.

Approach 1: Nodal Analysis



KCL at A:

$$10 \angle 0^\circ = \underline{I}_1 + \underline{I}_2 = \frac{V_A}{1 \Omega} + \frac{V_A}{(2 + j4) \Omega}$$

$$10 \angle 0^\circ = \frac{(2 + j4) \underline{V}_A + \underline{V}_A}{2 + j4} = \frac{3 + j4}{2 + j4} \underline{V}_A$$

$$\underline{V}_A = \left[\frac{2 + j4}{3 + j4} \right] \cdot 10 \angle 0^\circ$$

voltage divider to get $\underline{V}$:

$$\underline{V} = \underline{V}_A \left(\frac{2}{2 + j4} \right) = 10 \angle 0^\circ \left[\frac{2 + j4}{3 + j4} \right] \left[\frac{2}{2 + j4} \right]$$

↓ (continued)

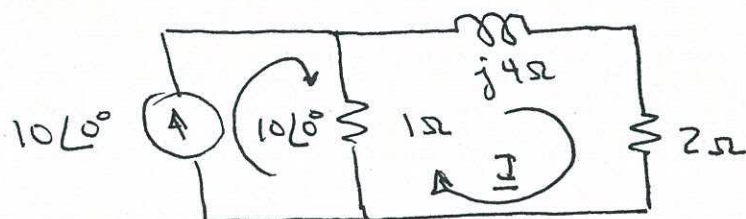
12.4 (cont'd)

Approach 1 (cont'd)

$$\underline{V} = 10 \angle 0^\circ \left[\frac{2}{3+j4} \right] = \frac{20 \angle 0^\circ}{\sqrt{3^2+4^2} \angle \tan^{-1}(4/3)}$$

$$\underline{V} = 4 \angle -53.3^\circ \Rightarrow \underline{v(t)} = \underline{4 \cos(8t - 53.3^\circ) \text{ V}}$$

Approach 2: Mesh analysis



$$1 \Omega [\underline{I} - 10 \angle 0^\circ] + j4 \Omega (\underline{I}) + 2 \underline{I} = 0$$

$$\underline{I} (j4 + 3) = 10 \angle 0^\circ$$

$$\underline{I} = \frac{10 \angle 0^\circ}{3 + j4} = \frac{10 \angle 0^\circ}{\sqrt{3^2+4^2} \angle \tan^{-1}(4/3)} = \frac{10 \angle 0^\circ}{5 \angle 53.13^\circ}$$

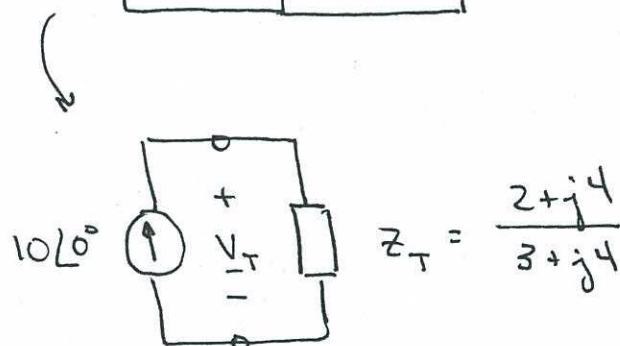
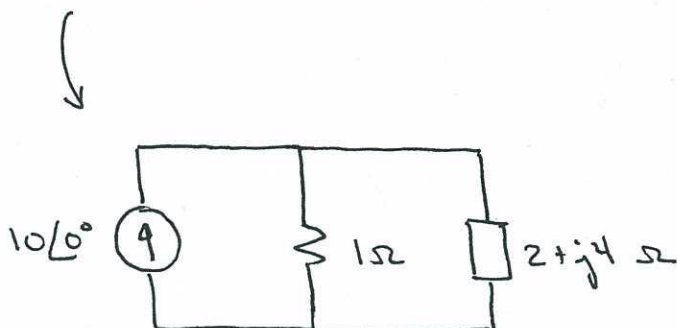
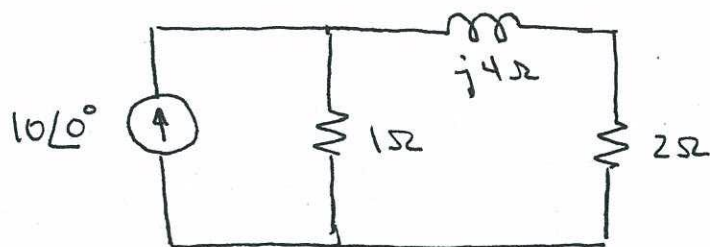
$$\underline{I} = 2 \angle -53.13^\circ \text{ A}$$

$$\underline{V} = (2 \Omega) \underline{I} = 4 \angle -53.13^\circ$$

$$\underline{v(t)} = \underline{4 \cos(8t - 53.13^\circ)}$$

12.4 (cont'd again...)

Approach 3: circuit reduction



$$\underline{V}_T = \underline{I}_s Z_T = 10\angle 0^\circ \left[\frac{2+j4}{3+j4} \right]$$

Voltage divider to get $\underline{V}$:

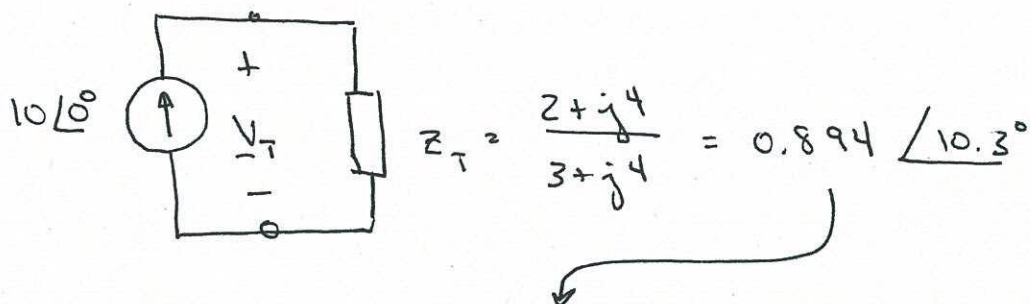
$$\underline{V} = \underline{V}_T \left(\frac{2}{2+j4} \right) = 10\angle 0^\circ \left[\frac{2+j4}{3+j4} \right] \left[\frac{2}{2+j4} \right]$$

$$\underline{V} = 4\angle -53.13^\circ$$

$$\underline{v(t)} = \underline{4 \cos(8t - 53.13^\circ)}$$

12.4 (still cont'd...)

(b) Use the equivalent circuit determined in Approach 3 above:



$$V_T = (10\angle 0^\circ)(0.894 \angle 10.3^\circ) = 8.944 \angle 10.29^\circ$$

$$P = \frac{V_m I_m}{2} \cos(\theta_r - \theta_i) = \frac{(10)(8.94)}{2} \cos(10.29^\circ - 0^\circ)$$

$$\underline{\underline{P = 44 \text{ W}}}$$

(c) Complex power delivered by source:

$$S = \frac{\underline{V}^* \underline{I}}{2} = \frac{(10\angle 0^\circ)(8.94 \angle 10.29^\circ)}{2} = 44.7 \angle 10.29^\circ$$

$$\text{Rectangular coordinates: } \underline{\underline{S = 44 + j8 \text{ VA}}}$$

(d) Inductor power:

$$\text{From approach 2 above, } \underline{I_1} = 2 \angle -53.13^\circ \text{ A}$$

$$Q = \frac{I^2}{2} [I_m(Z)] = \frac{2^2}{2} (4) = 8 \text{ VAR}$$

12.5

$$V_{RMS} = \frac{150,000 \text{ W}}{(0.7)(240 \text{ A})}$$

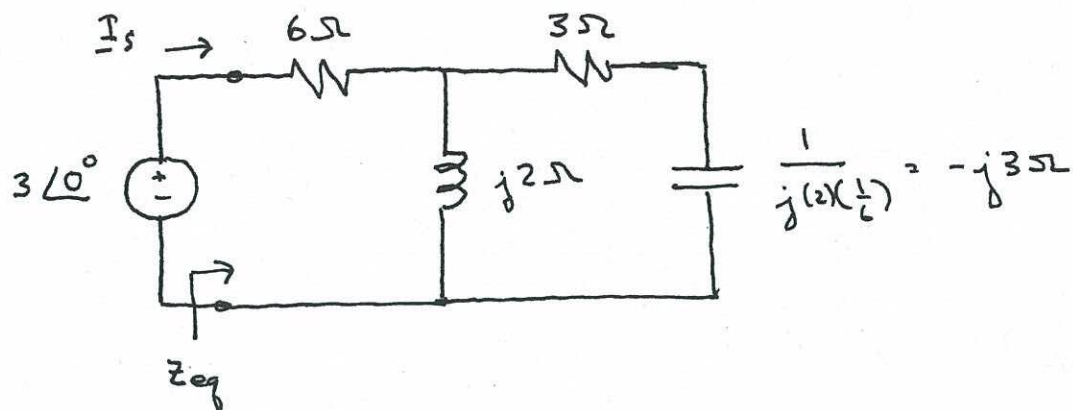
$$\underline{\underline{V_{RMS} = 892.9 \text{ V}}}$$

12.6

$$\theta = \cos^{-1}(0.8) = \underline{\underline{36.87^\circ}} \quad (\text{current } \underline{\text{lags}} \text{ voltage for inductive load})$$

12.7

Frequency domain circuit



(a) $S = \frac{V \cdot I_s^*}{2} \Rightarrow$ need to determine current out of source:

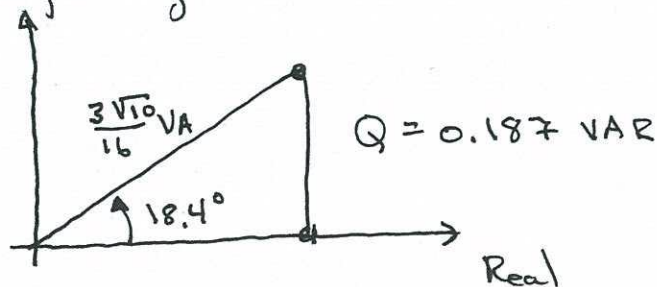
$$Z_{eq} = 6\Omega + \frac{(j2\Omega)(3-j3)\Omega}{j2\Omega + 3\Omega - j3\Omega} = 6\Omega + \frac{j6\Omega + 6\Omega}{(3-j)\Omega}$$

$$Z_{eq} = \frac{24}{3-j}$$

$$\underline{I}_s = \frac{3\angle 0^\circ}{Z_{eq}} = \frac{3\angle 0^\circ}{\frac{24}{3-j}} = \frac{3\sqrt{10}}{24} \angle -18.4^\circ$$

$$S = \frac{(3\angle 0^\circ) \left(\frac{3\sqrt{10}}{24} \angle +18.4^\circ \right)}{2} = \underline{\underline{\frac{3\sqrt{10}}{16} \angle 18.4^\circ}}$$

(b) Imaginary



$$P = 0.563 \text{ W}$$

12.7 continued

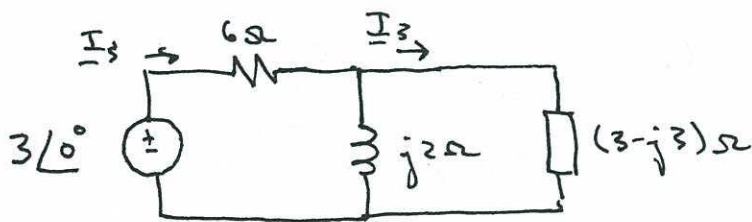
(c) Average power delivered by the source is the real part of the complex power (shown on part (b)).

$$P = \frac{3\sqrt{10}}{16} \cos(0 + 18.4^\circ) = \underline{\underline{0.563 \text{ W}}}$$

(d) The average power delivered by the source is dissipated in the resistors. Thus

$$P_{6\Omega} + P_{3\Omega} = P = \underline{\underline{0.563 \text{ W}}}$$

(e) Need to find current through the 3Ω resistor (or voltage difference across it) to get $P_{3\Omega}$. I'll use circuit reduction...

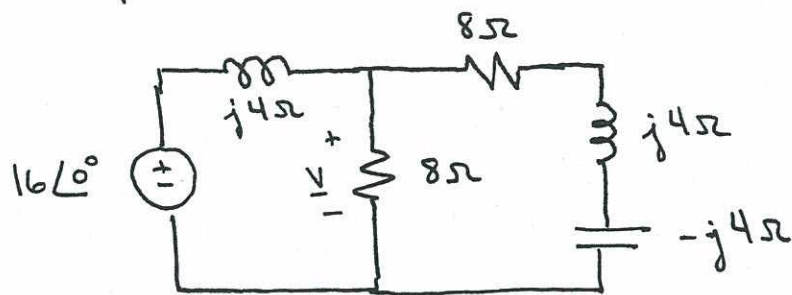


$$\begin{aligned} \underline{I}_3 &= \underline{I}_s \left[\frac{j2\Omega}{3-j3\Omega} \right] \\ &= \frac{3\angle 0^\circ (3-j)}{24} \left[\frac{j^2}{3-j} \right] \\ &= 0.25j = \frac{1}{4} \angle 90^\circ \end{aligned}$$

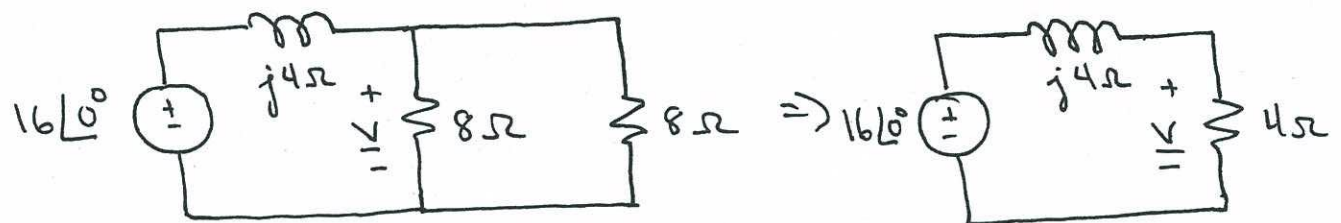
$$P_{3\Omega} = |\underline{I}_3|^2 \cdot (3\Omega) = \underline{\underline{\frac{3}{16} \text{ W}}}$$

12.8

Frequency domain circuit:

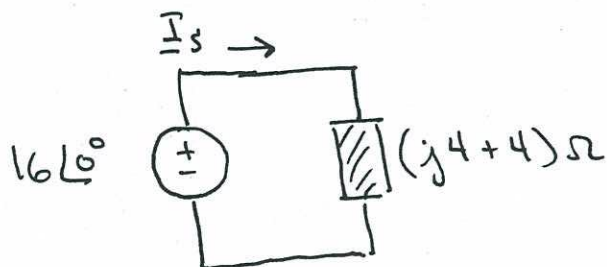


(a) Reduced circuit:



$$\underline{V} = 16\angle 0^\circ \left[\frac{4\Omega}{(4+j4)\Omega} \right] = \frac{16\angle 0^\circ}{\sqrt{2}\angle 45^\circ} = \frac{16}{\sqrt{2}} \angle -45^\circ$$

$$v(t) = \frac{16}{\sqrt{2}} \cos(2t - 45^\circ)$$

(b) Find source current, $\underline{I}_s$ 

$$\underline{I}_s = \frac{16\angle 0^\circ}{4+j4} = \frac{4}{\sqrt{2}} \angle -45^\circ$$

$$S' = \frac{\underline{V} \cdot \underline{I}^*}{2} = \frac{(16\angle 0^\circ) \left(\frac{4}{\sqrt{2}} \angle 45^\circ \right)}{2} = \frac{32}{\sqrt{2}} \angle 45^\circ$$

$$S = 16\sqrt{2} \angle 45^\circ$$

12.8 (cont'd)

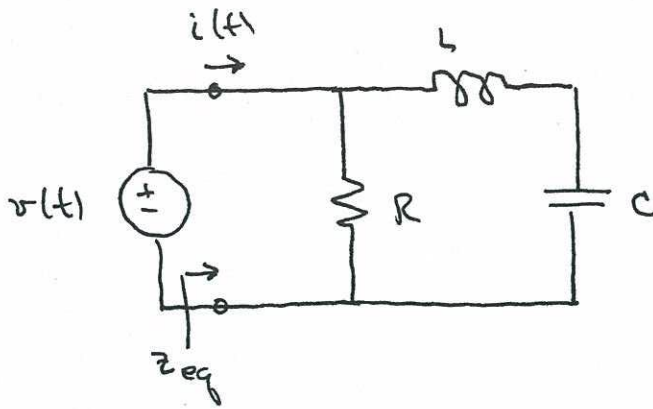
$$(c) \quad P = \operatorname{Re}\{s\} = |s| \cos(\theta_v - \theta_i)$$

$$\boxed{P = 16 \text{ W}}$$

$$(d) \quad Q = \operatorname{Im}\{s\} = |s| \sin(\theta_v - \theta_i)$$

$$\boxed{Q = 16 \text{ VAR}}$$

12.9



$$v(t) = 120 \cos(100t + 65^\circ) \Rightarrow \underline{V} = 120 \angle 65^\circ$$

$$i(t) = 2 \cos(100t + 30^\circ) \Rightarrow \underline{I} = 2 \angle 30^\circ$$

$$(a) \quad S' = \frac{\underline{V} \underline{I}^*}{2} = \frac{(120 \angle 65^\circ)(2 \angle -30^\circ)}{2} = \underline{\underline{120 \angle 35^\circ}}$$

(b) The power dissipated by the resistor is the same as the real power delivered by the source. Therefore,

$$P = |S| \cos(\theta_v - \theta_i) = 120 \cos(35^\circ)$$

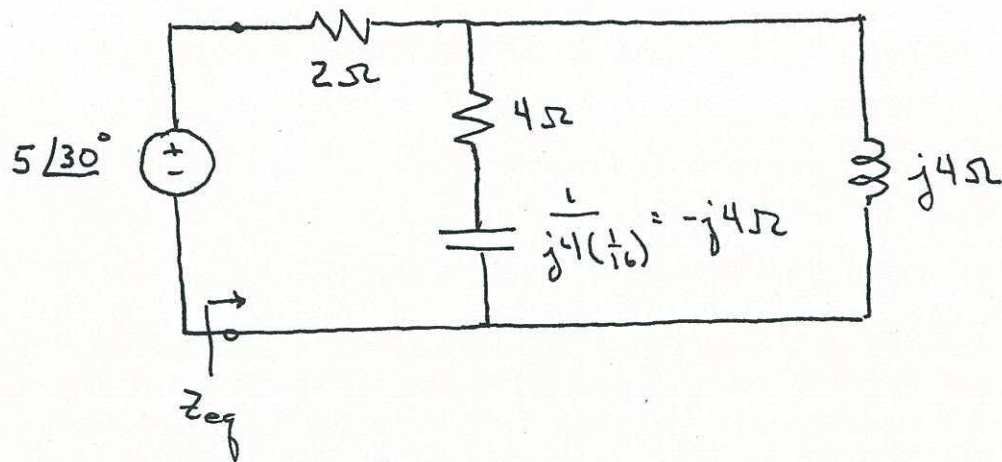
$$\boxed{P = 98.3 \text{ W}}$$

$$(c) \quad \underline{Z}_{eq} = \frac{\underline{V}}{\underline{I}} = \frac{120 \angle 65^\circ}{2 \angle 30^\circ}$$

$$\boxed{\underline{Z}_{eq} = 60 \angle 35^\circ}$$

12.10

Frequency domain circuit



(a) Circuit reduction:



$$z_{eq} = 2\Omega + \frac{j16 + 16}{4}\Omega$$

$$= (2 + j4 + 4)\Omega$$

$$z_{eq} = (6 + j4)\Omega$$

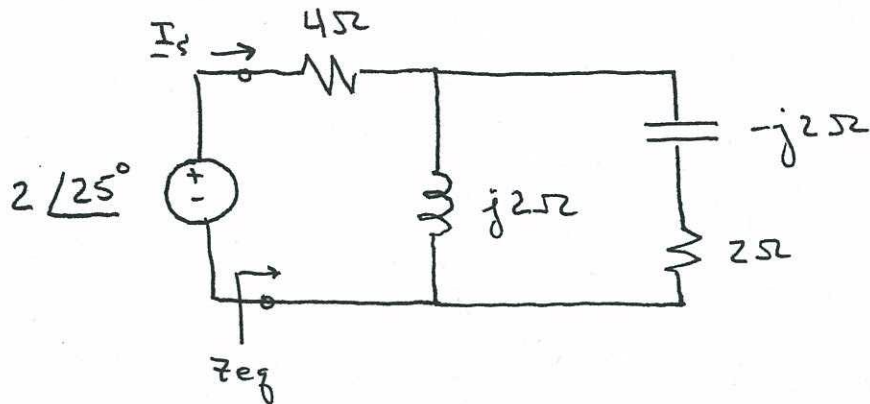
$$(b) \quad \underline{I}_s = \frac{5\angle 30^\circ}{6 + j4\Omega} = \frac{5}{2\sqrt{13}} \angle -3.7^\circ$$

$$S = \frac{\underline{V}_s \underline{I}_s^*}{2} = \frac{(5\angle 30^\circ) \left(\frac{5}{2\sqrt{13}} \angle +3.7^\circ \right)}{2} =$$

$$S = \frac{25}{2\sqrt{13}} \angle 33.7^\circ$$

12.11

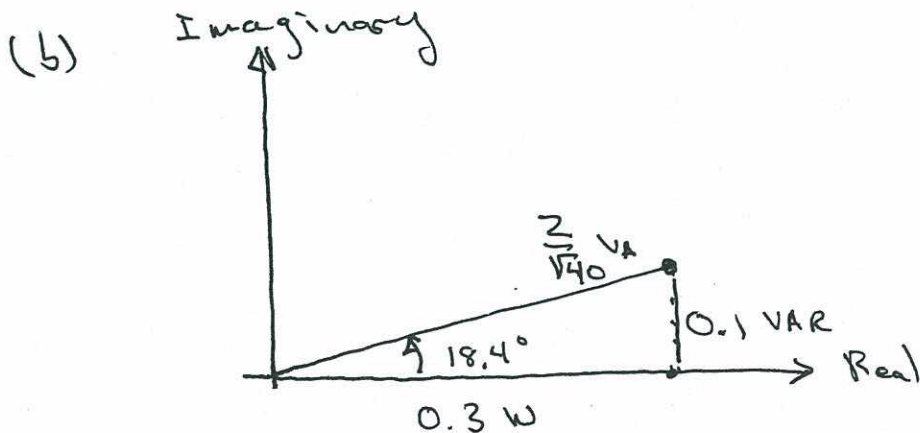
Frequency domain circuit



$$(a) \quad Z_{eq} = 4\Omega + \frac{(j2)(2-j2)}{j2+2-j2} = 4 + j2 + 2 = (6 + j2)\Omega$$

$$\underline{I}_s = \frac{2\angle 25^\circ}{6+j2} = \frac{2\angle 25^\circ}{\sqrt{40}\angle 18.4^\circ} = \frac{2}{\sqrt{40}}\angle 6.57^\circ$$

$$S = \frac{(2\angle 25^\circ)\left(\frac{2}{\sqrt{40}}\angle -6.57^\circ\right)}{2} \Rightarrow \boxed{\frac{2}{\sqrt{40}}\angle 18.43^\circ = S}$$



$$P = \frac{2}{\sqrt{40}} \cos(18.43^\circ) = 0.3 \text{ W}$$

$$Q = \frac{2}{\sqrt{40}} \sin(18.43^\circ) = 0.1$$